# Week 50: Staircases and lengths

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-50-facilitator.pdf, whose source is editable/facilitator-src/content.json with route-note.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the turn proof are in the guide.

## Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Two different staircases | P1 | P1 | P1 | first |
| A one-step and a two-step staircase | P2 | P2 | P2 | first |
| A staircase closer to the diagonal | P3 | P3 | P3 | first |
| Fewest turns inside the strip | P4 | P4 | P4 | first/return |
| Paths that may go left | P5 | P5 | | return |
| A leftward piece and less string | P6 | | | return |
| A narrower strip | | | P5 | return |
| The exact length of a path that goes left once | | | P6 | return |
| Could more, smaller pieces reach 200 mm? | | | P7 | return |

**First route.** K–1 P1–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P4. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P5; Grades 4–5 P4–P7.

## Key, one entry per problem

### Two different staircases

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

*Idea.* Every right-and-up path from S to F has length width + height = 280 mm.

### A one-step and a two-step staircase

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

*Idea.* Regrouping the right pieces and the up pieces shows the two lengths are equal.

### A staircase closer to the diagonal

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

*Idea.* Staircases can come as close to the diagonal as you like, yet every one stays 280 mm long; the diagonal is 200 mm.

### Fewest turns inside the strip

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · first/return route · wording differs by band

*Idea.* Inside the 15 mm strip the fewest turns is 8: a horizontal piece spans at most 40 mm, and 20 mm at either end.

## Paths that may go left

K–1 P5 · Grades 2–3 P5 · return route · wording differs by band

*Idea.* With total leftward travel L the length is 280 + 2L mm.

## A leftward piece and less string

K–1 P6 · return route

*Idea.* Every millimetre moved left must be made up by an extra millimetre right, so a leftward piece adds string.

## A narrower strip

Grades 4–5 P5 · return route

*Idea.* Halving the strip to 7.5 mm doubles the fewest turns to 16, and the staircase is still 280 mm.

## The exact length of a path that goes left once

Grades 4–5 P6 · return route

*Idea.* The right and left pieces still end 160 mm to the right, but together they travel more than 160 mm.

## Could more, smaller pieces reach 200 mm?

Grades 4–5 P7 · return route

*Idea.* Staircases approach the diagonal in position but not in length.